

KAPIL HINGU

United States | Email | (773) 200-9724 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineer with over **3 years of experience** designing, developing, and scaling high-performance web applications and cloud-native services. Combines a strong foundation in full-stack software development with practical expertise in modern AI and machine learning integrations, including Large Language Models (LLMs), RAG pipelines, and automated agentic workflows. Proven track record of optimizing backend architecture, reducing system latency, and delivering robust APIs in fast-paced production environments.

SKILLS

Core Languages & Web Development: Python, JavaScript ES6+, TypeScript, HTML5, CSS3, SQL, NoSQL, Web APIs

Backend Frameworks & Architecture: Node.js, Express.js, FastAPI, Django, Flask, RESTful APIs, GraphQL, Microservices Architecture, Event-Driven Architecture, ORM (SQLAlchemy, Prisma)

AI Software Engineering & GenAI: LLM Integration, Production RAG Architectures, Autonomous AI Agents, LangChain, LlamaIndex, OpenAI & Claude APIs, Prompt Engineering, Semantic Search & Embeddings, Function Calling, Tool Use,

Frontend Frameworks & UI: React.js, Next.js, React Native, Angular, Redux Toolkit, Context API, Tailwind CSS

Databases & Data Management: PostgreSQL, MongoDB, MySQL, SQL Server, Redis, Apache Kafka, Schema Design, Query Optimization, Database Migration Strategies, Caching Layers

Cloud Infrastructure & DevOps: AWS, EC2, S3, Lambda, CloudWatch, RDS, Azure, Google Cloud Platform, Docker, Kubernetes, Terraform, CI/CD Pipelines, GitHub Actions, Linux/Unix Administration, Nginx, Git/GitHub,

Testing, Security & Observability: PyTest, Jest, Mocha, Chai, Postman, Swagger, OpenAPI, Datadog, Prometheus, Grafana, OAuth2, JWT Authentication, OWASP Security Best Practices, CORS, Rate Limiting

EXPERIENCE

CVS Health, United States | *AI Software Engineer*

November 2025 – Present

- Engineered and deployed scalable **LLM-powered** clinical documentation processing pipelines utilizing **Python, FastAPI**, and **OpenAI APIs**, reducing manual review turnaround time by 35% across regional healthcare facilities.
- Architected a production-grade **Retrieval-Augmented Generation** system integrated with **Pinecone vector databases** and LangChain, improving medical knowledge retrieval precision and reducing hallucination rates by 28%.
- Developed autonomous **AI agents** using function calling and **semantic search** frameworks to automate patient triage workflows, handling over 15,000 monthly automated inquiries with a 94% resolution accuracy.
- Optimized microservices backend infrastructure and asynchronous **API endpoints** deployed on **AWS, Lambda, ECS** and Docker, reducing system latency by 42% under high-concurrency workloads.
- Implemented strict model evaluation protocols, **data masking**, and **JWT, OAuth2** security guardrails to ensure full HIPAA compliance across all generative AI applications and **cloud data storage** layers.

Reliance Jio Platforms, India | *Software Engineer*

July 2022 – July 2024

- Architected and scaled a high-traffic consumer web platform from 150K to 350K+ active users by engineering full-stack features end-to-end using **React, TypeScript, Java**, and **Node.js**, seamlessly processing 20M+ daily events.
- Developed robust payment microservices secured with **OAuth2** and **JWT authentication**, achieving zero critical production failures across high-volume monthly transaction workflows while implementing real-time status updates.
- Optimized heavy **PostgreSQL** backend workloads by cutting query execution times by 45% through strategic database schema redesign, advanced indexing, and performance tuning.
- Boosted core frontend web vitals, raising **Google Lighthouse performance** scores by 30% via component architecture refactoring, code-splitting, and building reusable React design systems adopted across the engineering team.
- Spearheaded quality assurance and **DevOps** efficiency by cutting sprint regression rates by 35% through automated testing pipelines, standardized **ESLint rules**, and streamlined CI/CD code review workflows within an 8-engineer Agile team.

Citus Infotech, India | *Software Engineer*

July 2021 – June 2022

- Built and optimized dynamic full-stack web modules using **Python, Django, Flask** and **JavaScript**, accelerating internal data processing workflows and reducing client-facing page load times by 30%.
- Designed and integrated **RESTful APIs** and **microservices** architectures to connect external third-party services, securely handling over 50,000 daily API requests with high availability.
- Refactored legacy SQL database schemas and implemented targeted query indexing in **PostgreSQL and MySQL**, lowering database response latency by 40% during peak operational traffic.
- Deployed containerized **microservices** environments using **Docker** and automated **CI/CD deployment** pipelines on **Linux servers**, cutting manual release deployment cycles by 50%.

EDUCATION

MS in Computer Science, DePaul University, Chicago, IL

September 2024 – June 2026

Relevant Coursework: Algorithms and Data Structures, Database Systems, Distributed Systems, Enterprise Computing, DevOps and Continuous Delivery, Artificial Intelligence, Machine Learning Applications, Agile Software Development

BE in Computer Engineering, Pillai HOC College of Engineering, India

June 2019 – June 2022

PROJECTS

OptiFi: AI Financial Intelligence Platform | [GitHub](#) | [Live Demo](#)

- Built a production multi agent AI platform end to end: LangGraph orchestration, RAG pipelines with pgvector semantic search, FastAPI REST APIs, React and TypeScript frontend, PostgreSQL on AWS Lambda with OAuth2 auth, GitHub Actions CI/CD, and CloudWatch monitoring.

AI Business Analyst | [GitHub](#) | [Live Demo](#)

- Built an AI-powered business analytics dashboard end to end with Next.js and Prisma, integrating Groq and OpenAI LLM APIs to generate datadriven insights, with NextAuth authentication and Recharts visualizations.